



Maharashtra Education Society's  
Institute of Management and Career Courses (IMCC), Pune  
AUTONOMOUS



**Academic Year 2026-27**  
**FY MCA- SEM I**  
**IT12 Data Structure and Algorithms**  
**Practice Assignment1**

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**Practical Assignment1-**

```
class Node:
    def __init__(self,value):
        self.data=value
        self.next=None

class SLL:
    def __init__(self):
        self.head=None

    #appending the node

    def append(self,new_node):
        if(self.head==None):
            self.head=new_node

        else:
            temp=self.head
            while temp.next:
                temp=temp.next
            temp.next=new_node

    #printing the list elements
    def print(self):
        temp=self.head
        while temp:
            print(temp.data)
            temp=temp.next
```



```
#inserting at specific position
def insert(self,new_node,pos):
    if(pos==1):
        new_node.next=self.head

        self.head=new_node

    else:
        p=1
        temp=self.head
        while(p!=pos-1):
            temp=temp.next
            p+=1
        new_node.next=temp.next
        temp.next=new_node

#deleting the node
def delete(self,value):
    temp=self.head
    if(temp.data==value):
        self.head=self.head.next
    else:
        while(temp.data!=value):
            prev=temp
            temp=temp.next
            if temp==None:
                print("value not present in the list")
                return

        prev.next=temp.next
        temp=None

#finding the middle number
def middleno(self):
    temp1=self.head
    temp2=self.head
```



```
while(temp2 and temp2.next):
    temp1=temp1.next
    temp2=temp2.next.next

if(temp1):
    print(temp1.data)
else:
    print("list is empty")

#reversing the list
def reverse(self):
    prev=None
    temp=self.head

    while temp:
        next_node=temp.next
        temp.next=prev
        prev=temp
        temp=next_node

    self.head=prev

#sum of two consecutive node values
def sumofcons(self):
    temp=self.head
    while temp and temp.next:
        total=temp.data+temp.next.data
        print(total)
        temp=temp.next
```



**#main program starts here**

**list1=SLL()**

**n1=Node(10)**

**n2=Node(20)**

**list1.append(n1)**

**list1.append(n2)**

**list1.print()**

**list1.insert(Node(30),2)**

**print()**

**list1.print()**

**print()**

**# list1.delete(20)**

**# print()**

**# list1.print()**

**list1.middleno()**

**print()**

**list1.reverse()**

**list1.print()**

**print()**

**list1.sumofcons()**



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**Output:**

```
Problems  Output  Debug Console  Terminal  Ports
PS D:\mesimcc> python linkedListprog.py
appended the node
10
20

node inserted at given position
10
30
20

list after node deletion
10
30

the middle no in list is
30

reverse list is
30
10

sum of consecutive node value
40
PS D:\mesimcc> 
```